Hang Yu

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Overview

My current research focuses on **Human-Centered AI** in **Robotics**, specifically, on learning from multi-modular teachings, including human feedback **(RLHF)**, demonstrations **(LfD)**, and generative methods**(VLM, VLA)**. He is now a research intern at ABB Robotics, working on VLAs for long-horizon tasks and **Agentic Robotics**.

Skills

Robotic Skills: ROS, ROS2, Gazebo, Isaac Sim, Gello, UMI, VLA, Sim2Real
Robotic Hardwares: Kinova Gen2/Gen3/Gen3-lite,ABB IRB 1300, Fetch, UR3E, UR5, Misty II, Cozmo
Machine Learning Skills: RL, Inverse RL, RLHF, LfD, RecSys, Imitation Learning, LLM/VLM/VLA Training

Education

Tufts University — Medford, MA
Ph.D. in Computer Science, GPA: 4.0/4.0 — Dec. 2026(expected)
Advisor: Dr. Elaine Schaertl Short

Tufts University — Medford, MA
M.S.E. in Computer Science, GPA: 3.95/4.0 — Jan. 2021
Advisor: Dr. Elaine Schaertl Short

Research/Work Experience

Research Intern, **ABB Robotics** — June 2026 – Present
- Build data effective VLAs for bimanual manipulation for long-horizon tasks.
- Exploring agentic robotics for industrial robot applications.

Research Scientist Intern, **Berkshire Grey,** **SoftBank Group** — Jan 2026 – June 2026
- Work on robot learning and developing VLA for bimanual-manipulation robotic arms.
- Build Gello and UMI pipeline for collecting data for VLA training and fine-tuning.

Research Assistant, Assistive Agent Behavior and Learning Lab, Tufts University — 2020 – present
- Conduct research on robot learning and human-in-the-loop learning.
- Robot learning from non-expert users with data collected in real-world scenarios in public spaces.

Publications

Peer-Reviewed Publications

C6 **Hang Yu**, James Staley, Shijie Fang, Jindan Huang, Wenchang Gao, Reuben M. Aronson, and E. Short, *PHIRL: Aligning Rewards with Progress for Inverse Reinforcement Learning with Vision-Language-Model-in-the-Loop*, (**CoRL**), 2026 (under review).

DC1 **Hang Yu**, Elaine Short, *Enabling Robust Learning from Non-Experts by Leveraging Human Demonstrations and Human Feedback.* IEEE International Conference on Robotics Automation **ICRA**, Doctoral Consortium, 2025, Atlanta, USA

C5 **Hang Yu**, Reuben M. Aronson, Katherine H. Allen, and E. Short, *From "Thumbs Up" to "10 out of 10": Reconsidering Scalar Feedback in Interactive Reinforcement Learning* 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), Detroit, USA, 2023.

C4 **Hang Yu**, Qidi Fang, Shijie Fang, Reuben M. Aronson, and E. Short, *How Much Progress Did I Make? An Unexplored Human Feedback Signal for Teaching Robots* 2024 IEEE **RO-MAN**, Pasadena, USA

C3 Matthew Ebisu*, **Hang Yu***, Reuben M. Aronson, Elaine S. Short, *See What I Mean? Expressiveness and Clarity in Robot Display Design*, IEEE **RO-MAN 2025**, Eindhoven, the Netherlands.

C2 Qidi Fang*, **Hang Yu***, Shijie Fang, Jindan Huang, Qiuyu Chen, Reuben M. Aronson, Elaine S. Short, *CHARM: Considering Human Attributes for Reinforcement Modeling*, IEEE **RO-MAN 2025**, Eindhoven, the Netherlands.

C1 Shijie Fang*, **Hang Yu***, Qidi Fang, Reuben M. Aronson, Elaine Schaertl Short, *Demonstration Sidetracks: Categorizing Systematic Non-Optimality in Human Demonstrations*, IEEE **RO-MAN 2025**, Eindhoven, the Netherlands.

(*) **means these authors contributed equally to the work.**

Journal Publications

J3 Tan, Z., **Yu, H.**, Wei, W., & Liu, J.. *Top-K interesting preference rules mining based on MaxClique.* **Expert Systems with Applications**, 143, 113043.

J2 **YU Hang**, WEI Wei, TAN Zheng, LIU Jing-lei. *Contextual Preference Collaborative Measure Framework Based on Belief System.* **Computer Science**, 2020, 47(4): 74-84.

J1 TAN, Z., LIU, J., & **YU, H.**. *Conditional preference mining based on MaxClique.* **Computer Applications**, 37(11), 3107.

Selected Workshop Papers

W2 **Hang Yu**, James Staley, Shijie Fang, Wenchang Gao, Reuben M. Aronson, and Elaine S. Short *PHIRL: Progress Heuristic for Inverse Reinforcement Learning.* RSS workshop 2025: Continual Robot Learning from Humans

W1 **Hang Yu** and Elaine Schaertl Short. 2021. *Active Feedback Learning with Rich Feedback.* HRI '21 Companion. Association for Computing Machinery, New York, NY, USA, 430–433.

Projects

Agentic Robotics
An agentic robotic skill for robots' self-training and improving in the loop.
Github: `https://github.com/HangYu8123/AgenticRobotics.git`

A World Model Powered VLA with Hierarchical Self-Attention for Manipulation
Integrating Hierarchical Self-Attention into GigaBrain-0 to achieve better grasping and object manipulation.
Github: `https://github.com/HangYu8123/gigabrain_with_Hierarchical_Self_Attention`

Harness Coding Workflow, GitHub 450+ Star★
Agentic Harness for looping, coding, debugging, refactoring, and logging.
Github: `https://github.com/HangYu8123/HarnessFlow.git`

PHIRL: Progress-Heuristicized Inverse Reinforcement Learning
An IRL learning framework that learns from human feedback, human demonstrations, and VLMs.
Github: `https://github.com/HangYu8123/PHIRL-Progress-Heuristic-for-Inverse-Reinforcement-Learning`

Honors/Awards

ICRA Doctoral Consortium 2025
Selected to participate in the ICRA Doctoral Consortium.

RO-MAN Student Grant 2024
Selected as the winner of the travel grant of RO-MAN 2024.

Lanqiao Programming Competition National First Prize 2016
Awarded to participants who were **Top 0.5%** among all the participants.

Selected Professional Service

Mentor
Isabella Bock, Undergraduate Student, Summer Intern, now at Tufts University
Matthew Ebisu, Master's Student, Co-author of C3, now at MassRobotics
Shijie Fang, Master's Student, Co-author of C4, C1, C2, now at Dongnan University

Teaching Assistant
Ethics for AI, Robotics, and Human-Robot Interaction, Tufts University — Spring 2024, Spring 2025
Human-Robot Interaction, Tufts University — Fall 2022

Selected Service
ICRA Undergraduate Research Panelist — 2025
Keynote Presenter at First Robotics Competition Conference NE — 2024
DIAMOND Program Panelist — 2023
Tufts Computer Science Student Council — 2023 – Present
Program Committee for AAMAS — 2026
Reviewer for International Conference on Robotics and Automation (ICRA) — 2023, 2024, 2025